



Started on	Thursday, 20 August 2026, 9:19 AM
State	Finished
Completed on	Thursday, 20 August 2026, 9:25 AM
Time taken	6 mins 34 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main()
3  {
4      int n,count=0;
5      scanf("%d",&n);
6      count++;
7      for(int i=n/2; i<n; i++)
8      {
9          count++;
10         for(int j=1; j<n; j = 2 * j)
11         {
12             count++;
13             for(int k=1; k<n; k = k * 2)
14             {
15                 count++;
16                 count++;
17             }
18             count++;
19         }
20         count++;
21     }
22     count++;
23     printf("%d",count);
24 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)